

Automated Testing Report

Inventory Management Project

Testing framework: pytest

Command used: python -m pytest app/tests -v

Test result:

28 tests collected

28 tests passed

0 failed

0 errors

Summary: The automated tests verify the core service logic of the updated inventory management system. The tests were adapted to the latest main branch structure and cover CategoryServices, LocationServices, ProductServices, UserService, and MovementService.

Test categories:

- Category service tests
- Location service tests
- Product service tests
- User service tests
- Movement service tests

Used test data: A clean in-memory SQLite database is created for each test using pytest fixtures. The base test data contains one category, one storage location, one admin user, and one product.

Final result: All 28 automated tests passed successfully.

Automated Test Case Overview

Test Case ID	Description	Expected Result	Actual Result	Status
TC_001	Create category successfully	Category is created successfully	Passed	Pass
TC_002	Create category without name raises error	ValueError is raised	Passed	Pass
TC_003	Get category successfully	Category is returned correctly	Passed	Pass
TC_004	Update category successfully	Category is updated correctly	Passed	Pass
TC_005	Delete category successfully	Category is deleted successfully	Passed	Pass
TC_006	Create location successfully	Location is created successfully	Passed	Pass
TC_007	Create location without name raises error	ValueError is raised	Passed	Pass

TC_008	Get location successfully	Location is returned correctly	Passed	Pass
TC_009	Update location successfully	Location is updated correctly	Passed	Pass
TC_010	Delete location successfully	Location is deleted successfully	Passed	Pass
TC_011	Create movement successfully	Movement is created successfully	Passed	Pass
TC_012	Create movement with invalid product raises error	ValueError is raised	Passed	Pass
TC_013	Create movement with invalid user raises error	ValueError is raised	Passed	Pass
TC_014	Create movement with invalid location raises error	ValueError is raised	Passed	Pass
TC_015	Get movement successfully	Movement is returned correctly	Passed	Pass
TC_016	Delete movement successfully	Movement is deleted successfully	Passed	Pass
TC_017	Move product to same location raises error	ValueError is raised	Passed	Pass
TC_018	Move product with insufficient stock raises error	ValueError is raised	Passed	Pass
TC_019	Create product successfully	Product is created successfully	Passed	Pass
TC_020	Create product with invalid category raises error	ValueError is raised	Passed	Pass
TC_021	Check product availability returns true	Product availability returns true	Passed	Pass
TC_022	Delete product successfully	Product is deleted successfully	Passed	Pass
TC_023	Get user successfully	User is returned correctly	Passed	Pass
TC_024	Get user not found raises error	ValueError is raised	Passed	Pass
TC_025	Login successfully	User login works correctly	Passed	Pass
TC_026	Login with wrong password raises error	ValueError is raised	Passed	Pass
TC_027	Change user role successfully	User role is updated correctly	Passed	Pass
TC_028	Change user role with invalid role raises error	ValueError is raised	Passed	Pass

Test Cases – Inventory Management Project

TC_001 – Create category successfully

Field	Details
Test case ID	TC_001
Test case title/description	Verify that a new category can be created successfully with valid input data.

Preconditions	A clean test database is available.
Test steps	1. Create CategoryServices. 2. Call create_category() with valid category data. 3. Save category in database. 4. Verify created category fields.
Test data/input	Category name: Electronics, type: sale
Expected result	Category is created successfully and stored in the database.
Actual result	Category was created successfully.
Status	Pass
Comments	No issues found.

TC_006 – Create location successfully

Field	Details
Test case ID	TC_006
Test case title/description	Verify that a new storage location can be created successfully with valid input data.
Preconditions	A clean test database is available.
Test steps	1. Create LocationServices. 2. Call create_location() with valid location data. 3. Save location in database. 4. Verify created location fields.
Test data/input	Location name: Main Warehouse, shelf number: A1
Expected result	Storage location is created successfully and stored in the database.
Actual result	Storage location was created successfully.
Status	Pass
Comments	No issues found.

TC_011 – Create movement successfully

Field	Details
Test case ID	TC_011
Test case title/description	Verify that a stock movement can be created successfully with valid product, user, and location data.
Preconditions	A clean test database is available. A product, user, and storage location already exist.
Test steps	1. Create MovementService. 2. Select existing product, user, and location from base test data. 3. Call create_movement() with valid movement data. 4. Verify created movement fields.
Test data/input	Product: Laptop, user: admin, location: Main Warehouse, quantity: 3, movement type: move
Expected result	A new stock movement record is created successfully in the database.
Actual result	Stock movement was created successfully.
Status	Pass
Comments	No issues found.

TC_017 – Move product to same location raises error

Field	Details
Test case ID	TC_017
Test case title/description	Verify that the system prevents moving a product to the same storage location.
Preconditions	A product and storage location already exist in the database.

Test steps	1. Create MovementService. 2. Select existing product and location. 3. Call move_product() using the same source and target location. 4. Check whether an error is raised.
Test data/input	Product: Laptop, source location: Main Warehouse, target location: Main Warehouse, quantity: 1
Expected result	The system raises ValueError with message “Source and target locations must be different”.
Actual result	ValueError was raised correctly.
Status	Pass
Comments	Business rule validation works correctly.

TC_019 – Create product successfully

Field	Details
Test case ID	TC_019
Test case title/description	Verify that a new product can be created successfully with valid input data.
Preconditions	A valid category and storage location already exist in the test database.
Test steps	1. Create ProductServices. 2. Call create_product() with valid product data. 3. Save product in database. 4. Verify created product fields.
Test data/input	Product name: Keyboard, quantity: 5, minimum stock: 1, category: Electronics
Expected result	Product is created successfully and stored in the database.
Actual result	Product was created successfully.
Status	Pass
Comments	No issues found.

TC_020 – Create product with invalid category raises error

Field	Details
Test case ID	TC_020
Test case title/description	Verify that the system prevents product creation with an invalid category ID.
Preconditions	ProductServices is available.
Test steps	1. Create ProductServices. 2. Call create_product() with invalid category ID. 3. Check whether an error is raised.
Test data/input	Invalid category id: 999
Expected result	The system raises ValueError with message “Category not found”.
Actual result	ValueError was raised correctly.
Status	Pass
Comments	Validation works correctly.

TC_025 – Login successfully

Field	Details
Test case ID	TC_025
Test case title/description	Verify that a user can log in successfully with correct credentials.
Preconditions	A valid user with hashed password exists in the database.

Test steps	1. Create UserService. 2. Call login() with valid username and password. 3. Verify returned user object.
Test data/input	Username: admin, password: password123
Expected result	User is authenticated successfully.
Actual result	Login completed successfully.
Status	Pass
Comments	Authentication works correctly.

TC_026 – Login with wrong password raises error

Field	Details
Test case ID	TC_026
Test case title/description	Verify that the system rejects login attempts with incorrect password.
Preconditions	A valid user exists in the database.
Test steps	1. Create UserService. 2. Call login() with incorrect password. 3. Check whether an error is raised.
Test data/input	Username: admin, password: wrongpassword
Expected result	The system raises ValueError with message “Incorrect password”.
Actual result	ValueError was raised correctly.
Status	Pass
Comments	Password validation works correctly.

Test Execution Result

```

PS C:\Users\aadat\Desktop\inventory-management-fhnw> python -m pytest app/tests -v
===== test session starts =====
platform win32 -- Python 3.13.7, pytest-9.0.3, pluggy-1.6.0 -- C:\Users\aadat\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\aadat\Desktop\inventory-management-fhnw
configfile: pytest.ini
plugins: anyio-4.12.1, cov-7.1.0
collected 28 items

app/tests/test_category_services.py::test_create_category_success PASSED [ 3%]
app/tests/test_category_services.py::test_create_category_without_name_raises_error PASSED [ 7%]
app/tests/test_category_services.py::test_get_category_success PASSED [ 10%]
app/tests/test_category_services.py::test_update_category_success PASSED [ 14%]
app/tests/test_category_services.py::test_delete_category_success PASSED [ 17%]
app/tests/test_location_services.py::test_create_location_success PASSED [ 21%]
app/tests/test_location_services.py::test_create_location_without_name_raises_error PASSED [ 25%]
app/tests/test_location_services.py::test_get_location_success PASSED [ 28%]
app/tests/test_location_services.py::test_update_location_success PASSED [ 32%]
app/tests/test_location_services.py::test_delete_location_success PASSED [ 35%]
app/tests/test_movement_services.py::test_create_movement_success PASSED [ 39%]
app/tests/test_movement_services.py::test_create_movement_with_invalid_product_raises_error PASSED [ 42%]
app/tests/test_movement_services.py::test_create_movement_with_invalid_user_raises_error PASSED [ 46%]
app/tests/test_movement_services.py::test_create_movement_with_invalid_location_raises_error PASSED [ 50%]
app/tests/test_movement_services.py::test_get_movement_success PASSED [ 53%]
app/tests/test_movement_services.py::test_delete_movement_success PASSED [ 57%]
app/tests/test_movement_services.py::test_move_product_same_location_raises_error PASSED [ 60%]
app/tests/test_movement_services.py::test_move_product_not_enough_stock_raises_error PASSED [ 64%]
app/tests/test_product_services.py::test_create_product_success PASSED [ 67%]
app/tests/test_product_services.py::test_create_product_with_invalid_category_raises_error PASSED [ 71%]
app/tests/test_product_services.py::test_check_availability_returns_true PASSED [ 75%]
app/tests/test_product_services.py::test_delete_product_success PASSED [ 78%]
app/tests/test_user_service.py::test_get_user_success PASSED [ 82%]
app/tests/test_user_service.py::test_get_user_not_found_raises_error PASSED [ 85%]
app/tests/test_user_service.py::test_login_success PASSED [ 89%]
app/tests/test_user_service.py::test_login_wrong_password_raises_error PASSED [ 92%]
app/tests/test_user_service.py::test_change_role_success PASSED [ 96%]
app/tests/test_user_service.py::test_change_role_invalid_role_raises_error PASSED [100%]

===== 28 passed in 0.67s =====
PS C:\Users\aadat\Desktop\inventory-management-fhnw>

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All automated tests were executed successfully using pytest.